Pearson Edexcel International GCSE

Mathematics A

Modular Unit 2

November 2023 Reconstructed Past Paper

DETAILED SOLUTIONS

Original question followed by a full worked solution

36 adaptive question sections 113 marks

Selected from Linear Higher Papers 1H and 2H

Dr Eslam Ahmed

Elite IGCSE Mathematics • eliteigcse.com

L-to-M | CONVERTED MODULAR EDITION

Selected from Linear Higher papers; not an original Modular examination paper.

Assistant Lecturer, Faculty of Engineering, Cairo University

B.Sc. Electrical Engineering (2017), Distinction with Honours
M.Sc. Mathematics (2023)
PhD Candidate

eliteigcse.com • WhatsApp +20 112 000 9622

About this reconstructed paper

Each question section appears in the same order as the Student book: the complete source demand is followed by its demand-local worked solution and final answer.

Ownership rule: Unit 2 owns a demand when its assessed or terminal statement is Unit 2, even if Unit 1 skills are prerequisites. For example, drawing boundary lines supports the Unit 2 region-of-inequalities demand. This book contains the demands owned by Unit 2; the selection contains 113 marks.

This is an Elite reconstructed teaching paper selected from official Linear Higher material; it is not an official Pearson Modular examination paper. Source assets are retained for private teaching use.

Contents

Question 01

4MA1-1H Q3 demand

ORIGINAL QUESTION UNIT 2

4 marks

3 Joshua buys a car for \$12 500

He sells the car to Nina.

Nina pays

- a deposit of \$1500
- followed by 36 monthly payments of \$450

Work out Joshua's percentage profit.

.... %

Worked solution 01

Percentage profit

MODULAR UNIT 2
4 marks

Demand – Percentage profit

Find revenue

$$1500 + 36(450) = 17700.$$

Find profit

$$17700 - 12500 = 5200.$$

Convert to a percentage

$$\frac{5200}{12500} \times 100 = 41.6\%.$$

Final answer

41.6%

Question 02

4MA1-1H Q6 demand

ORIGINAL QUESTION UNIT 2

4 marks

6 Juan wants to buy a ticket to fly from Madrid to Berlin.

He finds two different types of ticket he can buy in a sale, ticket **A** and ticket **B**

ticket **A**	ticket **B**
$\frac{1}{6}$ off normal price	20% off normal price

The sale price of ticket **A** is 140 euros.
The sale price of ticket **B** is 136 euros.

Work out the difference between the normal price of ticket **A** and the normal price of ticket **B**

.... euros

Worked solution 02

Recover normal ticket prices

MODULAR UNIT 2

4 marks

Demand – Recover normal ticket prices

Ticket A

$$\frac{5}{6}N_A = 140 \Rightarrow N_A = 168.$$

Ticket B

$$0.8N_B = 136 \Rightarrow N_B = 170.$$

Compare

$$170 - 168 = 2.$$

Final answer

2 euros

Question 03

4MA1-1H Q7 demand

ORIGINAL QUESTION UNIT 2

2 marks

7 $A = 5^3 \times 7^3 \times 11^6$ and $B = 5^6 \times 7^2 \times 11^4$

Find the highest common factor (HCF) of A and B
Give your answer as a product of powers of its prime factors.

....

Worked solution 03

Highest common factor

MODULAR UNIT 2
2 marks

Demand – Highest common factor

Take minimum exponents

$$\text{HCF} = 5^3 \times 7^2 \times 11^4.$$

Final answer

5^3 x 7^2 x 11^4

Question 04

4MA1-1H Q8 (a)

ORIGINAL QUESTION UNIT 2

2 marks

8 (a) Solve the inequality $8x - 4 \geqslant 3x - 10$

....

(2)

Worked solution 04

Solve the linear inequality

MODULAR UNIT 2

2 marks

(a) – Solve the linear inequality

Collect terms

$$8x - 4 \geq 3x - 10 \Rightarrow 5x \geq -6.$$

Divide by positive 5

$$x \geq -\frac{6}{5}.$$

Final answer

x ≥ -6/5

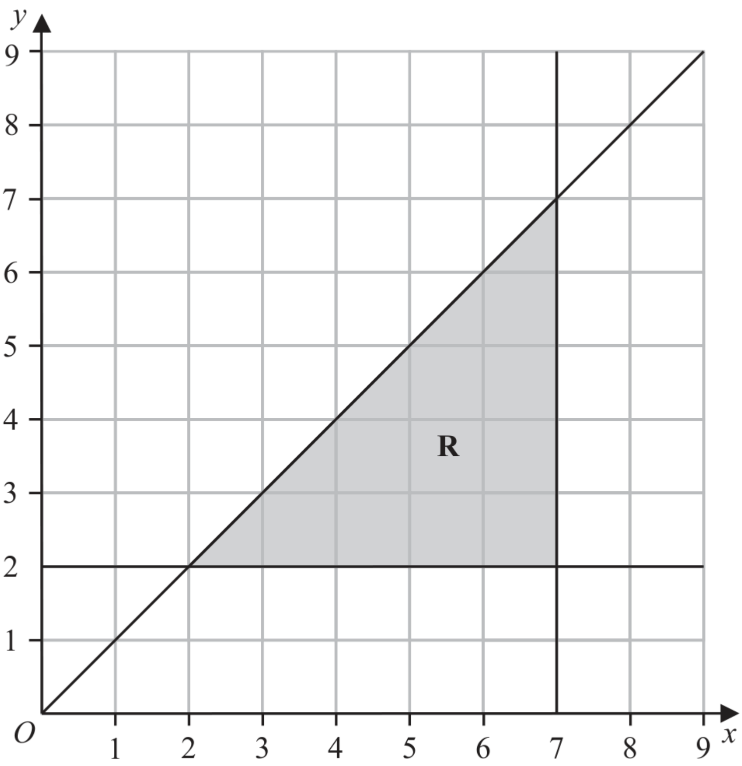
Question 05

4MA1-1H Q8 (b)

ORIGINAL QUESTION UNIT 2

3 marks

The region **R**, shown shaded in the diagram, is bounded by three straight lines.



(b) Write down the three inequalities that define the region **R**

....

....

....

(3)

Worked solution 05

Define the shaded region

MODULAR UNIT 2
3 marks

(b) – Define the shaded region

Read each boundary

$$y = 2, \quad x = 7, \quad y = x.$$

Choose the shaded side

$$R: \ y \geq 2, \ x \leq 7, \ y \leq x.$$

Final answer

$y \geq 2, x \leq 7, y \leq x$

Question 06

4MA1-1H Q9 (a)

ORIGINAL QUESTION UNIT 2

1 marks

9 (a) Write 5.87×10^{-4} as an ordinary number.

....

(1)

Worked solution 06

Ordinary number

MODULAR UNIT 2
1 marks

(a) – Ordinary number

Move the decimal four places left

$$5.87 \times 10^{-4} = 0.000587.$$

Final answer

0.000587

Question 07
4MA1-1H Q9 (b)

ORIGINAL QUESTION UNIT 2
1 marks

(b) Write 84 000 000 in standard form.

....
(1)

Worked solution 07

Standard form

MODULAR UNIT 2

1 marks

(b) – Standard form

Write one non-zero digit first

$$84000000 = 8.4 \times 10^7.$$

Final answer

8.4 x 10^7

Question 08

4MA1-1H Q9 (c)

ORIGINAL QUESTION UNIT 2

2 marks

The number of neurons in a human brain is 8.5×10^{10}
The number of neurons in a monkey brain is 1.47×10^{9}

The number of neurons in a human brain is $K \times$ the number of neurons in a monkey brain.

(c) Work out the value of K
Give your answer correct to one decimal place.

$K =$ ....

(2)

Worked solution 08

Standard-form ratio

MODULAR UNIT 2

2 marks

(c) – Standard-form ratio

Form the ratio

$$K = \frac{8.5 \times 10^{10}}{1.47 \times 10^{9}} = \frac{85}{1.47}.$$

Evaluate

$$K = 57.823... \approx 57.8.$$

Final answer

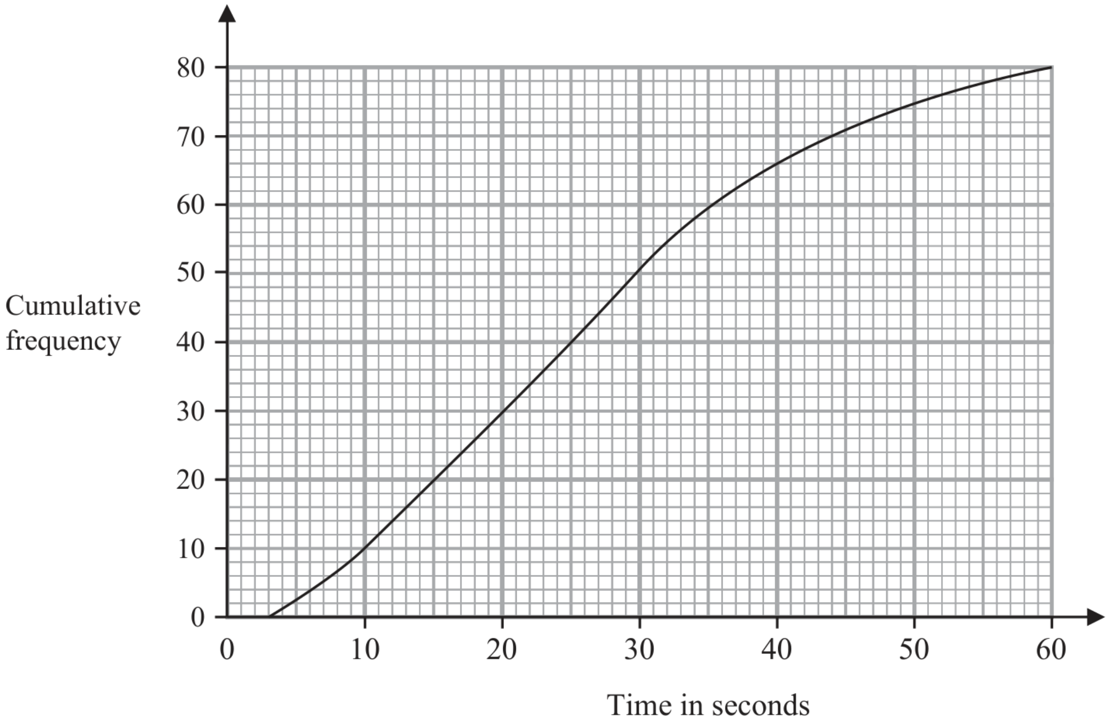
K = 57.8

Question 09

4MA1-1H Q13 (a)

ORIGINAL QUESTION UNIT 2
1 marks

13 The cumulative frequency graph gives information about the times, in seconds, that 80 adults took to log in to an online bank.



(a) Find an estimate for the median time.

.... seconds

(1)

Worked solution 09

Median from cumulative frequency

MODULAR UNIT 2
1 marks

(a) – Median from cumulative frequency

Read the middle rank

$$80/2 = 40; reading CF = 40 from the graph gives about 25 seconds.$$

Final answer

25 seconds

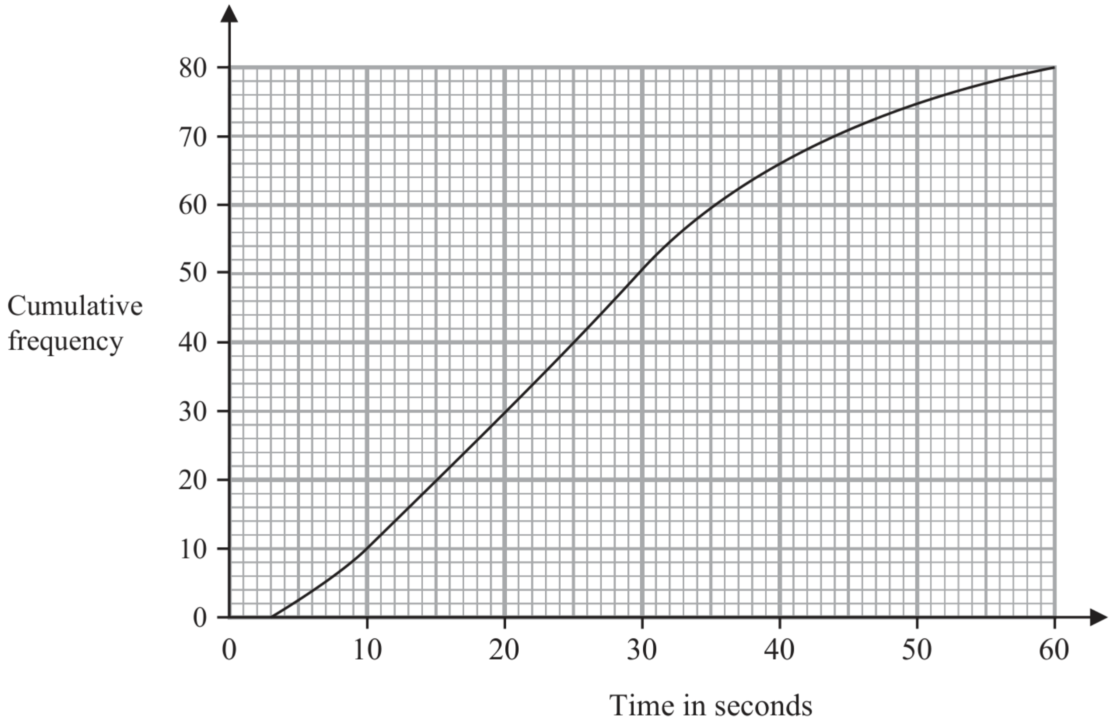
Question 10

4MA1-1H Q13 (b)

ORIGINAL QUESTION UNIT 2

3 marks

13 The cumulative frequency graph gives information about the times, in seconds, that 80 adults took to log in to an online bank.



(b) Work out the percentage of these adults that took longer than 50 seconds to log in.
Show your working clearly.

....%

(3)

Worked solution 10

Percentage taking longer than 50 seconds

MODULAR UNIT 2
3 marks

(b) – Percentage taking longer than 50 seconds

Read CF at 50

$$CF(50) \approx 75, \quad 80 - 75 = 5.$$

Convert to a percentage

$$\frac{5}{80} \times 100 = 6.25\%.$$

Final answer

6.25%

Question 11

4MA1-1H Q14 demand

ORIGINAL QUESTION UNIT 2

3 marks

14

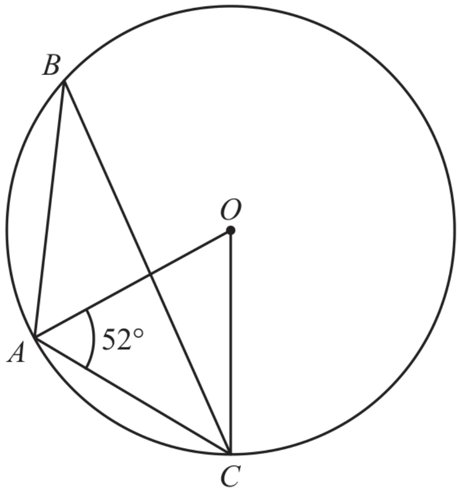


Diagram **NOT** accurately drawn

A, B and C are points on a circle, centre O

Angle $OAC = 52°$

Find the size of angle ABC
Give reasons for your working.

.... °

Worked solution 11

Angle at the circumference

MODULAR UNIT 2
3 marks

Demand – Angle at the circumference

Find the central angle

$$\angle AOC = 180^\circ - 2(52^\circ) = 76^\circ.$$

Use the circle theorem

$$\angle ABC = \frac{1}{2}\angle AOC = 38^\circ.$$

State reasons

$$OA = OC(radii), so base angles are equal; angle at centre is twice angle at circumference.$$

Final answer

38°

Question 12

4MA1-1H Q15 demand

ORIGINAL QUESTION UNIT 2

3 marks

15 Make n the subject of the formula $x = \dfrac{3p + n}{3n - 4}$

....

Worked solution 12

Make n the subject

MODULAR UNIT 2
3 marks

Demand – Make n the subject

Clear the denominator

$$x(3n-4)=3p+n \Rightarrow 3nx-4x=3p+n.$$

Collect n terms

$$n(3x-1)=3p+4x.$$

Divide

$$n=\frac{3p+4x}{3x-1}.$$

Final answer

n = (3p + 4x)/(3x - 1)

Question 13

4MA1-1H Q16 demand

ORIGINAL QUESTION UNIT 2

4 marks

16 A curve has equation $y = 4x^3 - 8x + 5$

Find the x coordinates of the two points on the curve where the gradient is $\frac{1}{3}$

....

Worked solution 13

Coordinates with a given gradient

MODULAR UNIT 2

4 marks

Demand – Coordinates with a given gradient

Differentiate

$$\frac{dy}{dx} = 12x^2 - 8.$$

Set the gradient

$$12x^2 - 8 = \frac{1}{3} \Rightarrow 12x^2 = \frac{25}{3} \Rightarrow x^2 = \frac{25}{36}.$$

Take both roots

$$x = \pm\frac{5}{6}.$$

Final answer

x = ±5/6

Question 14

4MA1-1H Q18 demand

ORIGINAL QUESTION UNIT 2
5 marks

18 The diagram shows the positions of three villages, A, B and C

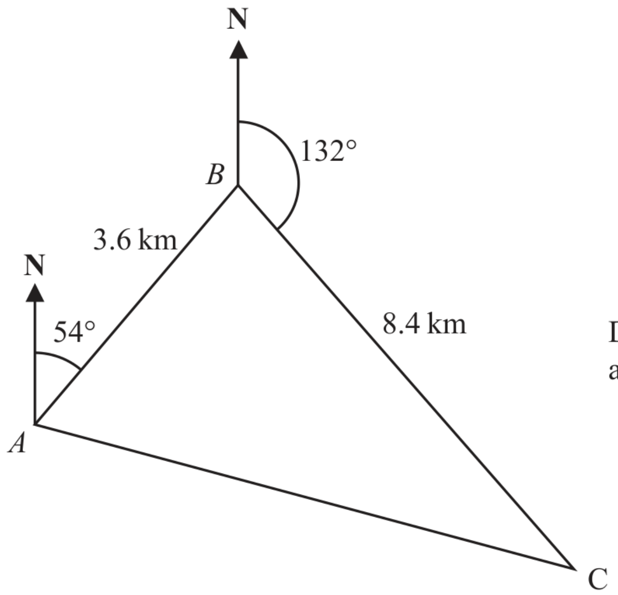


Diagram **NOT** accurately drawn

The bearing of B from A is 054°
The bearing of C from B is 132°

Melur walks from A to B
She then walks from B to C and from C to A

Melur walks at an average speed of 6 km/h

Work out the total time Melur takes.
Give your answer in hours and minutes.

.... hours .... minutes

Worked solution 14

Bearings and total walking time

MODULAR UNIT 2
5 marks

Demand – Bearings and total walking time

Find angle ABC

$$\angle ABC = 54^\circ + (180^\circ - 132^\circ) = 102^\circ.$$

Find AC

$$AC^2 = 3.6^2 + 8.4^2 - 2(3.6)(8.4)\cos 102^\circ \Rightarrow AC \approx 9.802 \text{ km}.$$

Find time

$$t = \frac{3.6 + 8.4 + 9.802}{6} = 3.6337 \text{ h} = 3 \text{ h } 38 \text{ min (approximately)}.$$

Final answer

3 hours 38 minutes

Question 15

4MA1-1H Q19 demand

ORIGINAL QUESTION UNIT 2

6 marks

19 Here are the first 4 terms in an arithmetic sequence.

3 7 11 15

The last term of the sequence is x
The sum of the terms of the sequence is 7260

Find the value of x
Show clear algebraic working.

$x =$ ....

Worked solution 15

Last term of an arithmetic sequence

MODULAR UNIT 2
6 marks

Demand – Last term of an arithmetic sequence

Identify a and d

$$a = 3, \quad d = 4.$$

Use the sum

$$S_n = \frac{n}{2}[2(3) + (n-1)4] = 7260 \Rightarrow 2n^2 + n - 7260 = 0.$$

Solve for the number of terms

$$(2n + 121)(n - 60) = 0 \Rightarrow n = 60.$$

Find the last term

$$x = 3 + (60 - 1)4 = 239.$$

Final answer

239

Question 16

4MA1-1H Q21 demand

ORIGINAL QUESTION UNIT 2

5 marks

21 Here is a triangular prism *ABCDEF*

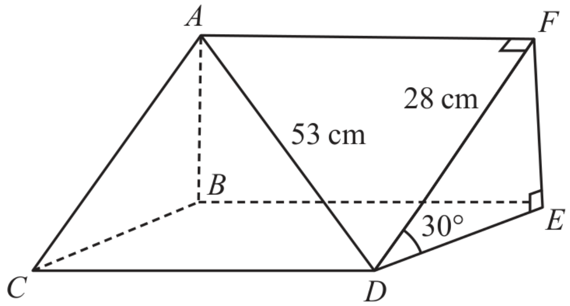


Diagram **NOT** accurately drawn

$AD = 53$ cm
$DF = 28$ cm
Angle $FDE = 30°$

Work out the volume of the triangular prism.
Give your answer correct to the nearest whole number.

.... cm^3

Worked solution 16

Volume of a triangular prism

MODULAR UNIT 2

5 marks

Demand – Volume of a triangular prism

Resolve the triangular end

$$FE = 28\sin 30^\circ = 14, \quad DE = 28\cos 30^\circ = 14\sqrt{3}.$$

Find prism length

$$AF = \sqrt{53^2 - 28^2} = 45.$$

Calculate volume

$$V = \frac{1}{2}(14)(14\sqrt{3})(45) = 4410\sqrt{3} \approx 7638.34.$$

Final answer

$7638\ \mathrm{cm}^3$

Question 17

4MA1-1H Q22 demand

ORIGINAL QUESTION UNIT 2

6 marks

22 [In this question 1 cm = 1 unit on the x-axis and 1 cm = 1 unit on the y-axis]

P is a point on a circle with centre (0, 0)
The coordinates of P are (8, –10)

The line **L** is the tangent to the circle at the point P
L crosses the x-axis at the point Q and crosses the y-axis at the point R

Work out the length of RQ
Give your answer correct to 3 significant figures.

.... cm

Worked solution 17

Length between tangent intercepts

MODULAR UNIT 2

6 marks

Demand – Length between tangent intercepts

Find radius gradient

$$m_{OP} = (-10 - 0)/(8 - 0) = -5/4.$$

Find tangent gradient

$$m_L = 4/5 \text{ since } m_{OP} m_L = -1.$$

Find intercepts

$$y + 10 = \frac{4}{5}(x - 8) \implies R = (0, -16.4),\ Q = (20.5, 0).$$

Distance

$$RQ = \sqrt{20.5^2 + 16.4^2} = 26.3 \text{ cm (3 s.f.)}.$$

Final answer

26.3 cm

Question 18

4MA1-1H Q23 demand

ORIGINAL QUESTION UNIT 2
5 marks

23 Solid **A** is similar to solid **B**

Here is some information about solid **A** and solid **B**

	solid **A**	solid **B**
Height (cm)	3^x	
Area (cm^2)	7776	486
Volume (cm^3)	8^x	2^{x+4}

Work out the height of solid **B**
Give your answer as a decimal.

.... cm

Worked solution 18

Similar solids: height of B

MODULAR UNIT 2
5 marks

Demand – Similar solids: height of B

Find the linear scale factor

$$\frac{A_A}{A_B} = \frac{7776}{486} = 16 \Rightarrow k = 4.$$

Use the volume ratio

$$\frac{8^x}{2^{x+4}} = k^3 = 64 \Rightarrow 2^{2x-4} = 2^6 \Rightarrow x = 5.$$

Scale the height

$$h_A = 3^5 = 243, \quad h_B = 243/4 = 60.75 \text{ cm}.$$

Final answer

60.75 cm

Question 19

4MA1-1H Q24 (a)

ORIGINAL QUESTION UNIT 2
2 marks

24 The curve with equation $f(x) = 5x^2 + 9x + 2$ is transformed to the curve with equation

$g(x) = 5(x+4)^2 + 9(x+4) + 8$ by the translation $\begin{pmatrix} a \\ b \end{pmatrix}$

(a) Write down the value of a and the value of b

$a =$ ....

$b =$ ....

(2)

Worked solution 19

Translation of a function

MODULAR UNIT 2

2 marks

(a) – Translation of a function

Match the x shift

$$g(x) = f(x+4) + 6, \text{ so the graph moves 4 left and 6 up.}$$

Final answer

(a,b) = (-4, 6)

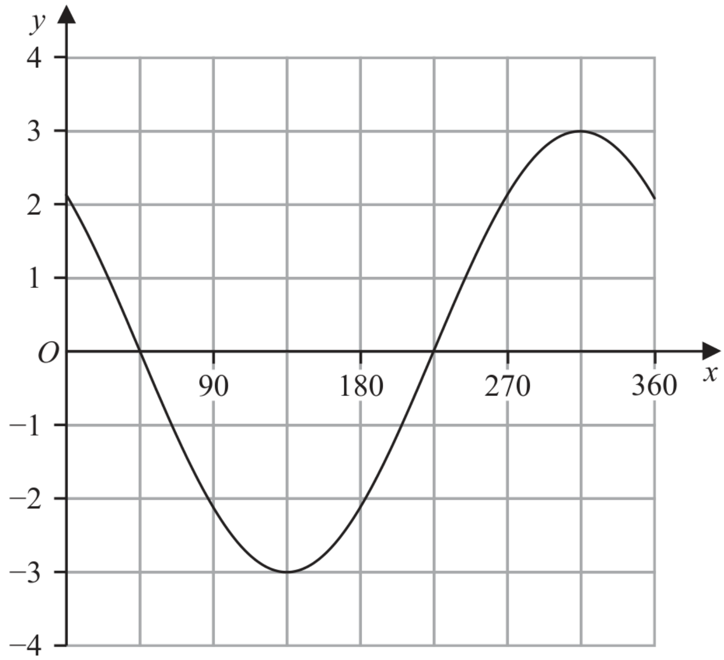
Question 20

4MA1-1H Q24 (b)

ORIGINAL QUESTION UNIT 2

2 marks

The graph of $y = p\cos(x+q)^\circ$ for $0 \leqslant x \leqslant 360$ is drawn on the grid below.



Given that $p > 0$ and $0 < q < 360$

(b) find the value of p and the value of q

$p =$ ....

$q =$ ....

(2)

Worked solution 20

Read p and q from a cosine graph

MODULAR UNIT 2
2 marks

(b) – Read p and q from a cosine graph

Read the amplitude

$$\max y = 3, \quad p = 3\ (p > 0).$$

Read the phase shift

$$The maximum occurs at x = 315^\circ;\ 315 + q = 360 \Rightarrow q = 45^\circ.$$

Final answer

p = 3, q = 45

Question 21
4MA1-2H Q1 (a)

ORIGINAL QUESTION UNIT 2
1 marks

1 The table shows information about the lengths, in minutes, of 50 telephone calls.

Length of telephone call (m minutes)	**Frequency**
$0 < m \leqslant 5$	8
$5 < m \leqslant 10$	2
$10 < m \leqslant 15$	6
$15 < m \leqslant 20$	4
$20 < m \leqslant 25$	12
$25 < m \leqslant 30$	18

(a) Write down the modal class.

....

(1)

Worked solution 21

Modal class

MODULAR UNIT 2

1 marks

(a) – Modal class

Read the greatest frequency

The greatest frequency is 18, in the class $25 < m \leq 30$.

Final answer

25 < m <= 30

Question 22

4MA1-2H Q1 (b)

ORIGINAL QUESTION UNIT 2

3 marks

1 The table shows information about the lengths, in minutes, of 50 telephone calls.

Length of telephone call (m minutes)	**Frequency**
$0 < m \leqslant 5$	8
$5 < m \leqslant 10$	2
$10 < m \leqslant 15$	6
$15 < m \leqslant 20$	4
$20 < m \leqslant 25$	12
$25 < m \leqslant 30$	18

(b) Work out an estimate for the total length, in minutes, of these telephone calls.

.... minutes

(3)

Worked solution 22

Estimated total length from grouped data

MODULAR UNIT 2
3 marks

(b) – Estimated total length from grouped data

Use class midpoints

$$2.5(8) + 7.5(2) + 12.5(6) + 17.5(4) + 22.5(12) + 27.5(18).$$

Add the midpoint products

$$20 + 15 + 75 + 70 + 270 + 495 = 945.$$

Final answer

945 minutes

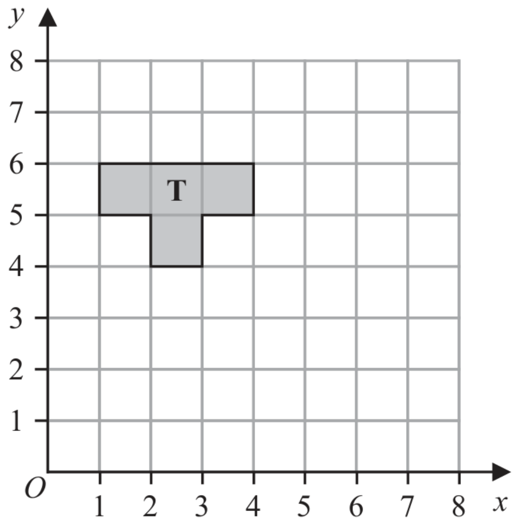
Question 23

4MA1-2H Q3 (a)

ORIGINAL QUESTION UNIT 2

2 marks

3



(a) Reflect shape **T** in the line $y = x$

(2)

Worked solution 23

Reflection in y = x

MODULAR UNIT 2

2 marks

(a) – Reflection in y = x

Apply the reflection rule

$$\text{Reflection in } y = x \text{ maps } (x, y) \mapsto (y, x).$$

Plot the image

$$\text{The image vertices are } (4, 2), (4, 3), (5, 3), (5, 4), (6, 4), (6, 1), (5, 1), (5, 2);\ \text{join them in order.}$$

Final answer

Reflected shape with vertices (4,2), (4,3), (5,3), (5,4), (6,4), (6,1), (5,1), (5,2)

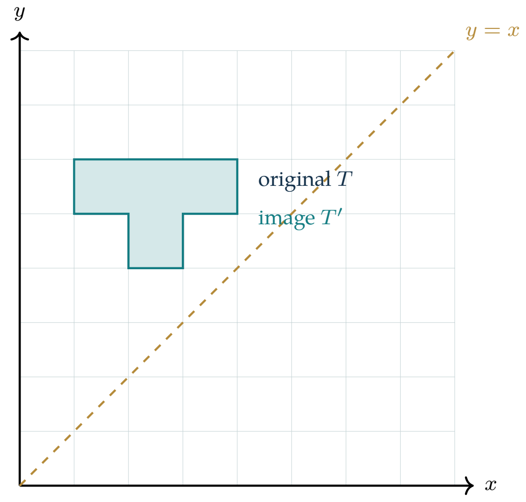
Completed diagram 23

Reflection in y = x

MODULAR UNIT 2

2 marks

Completed solution diagram: Reflection in $y = x$

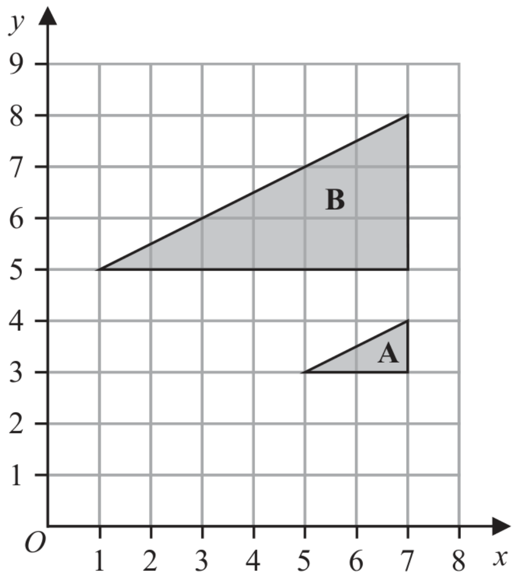


Question 24

4MA1-2H Q3 (b)

ORIGINAL QUESTION UNIT 2

3 marks



(b) Describe fully the single transformation that maps triangle **A** onto triangle **B**

....

....

(3)

Worked solution 24

Describe the enlargement

MODULAR UNIT 2
3 marks

(b) – Describe the enlargement

Compare corresponding lengths

Every side of B is three times the corresponding side of A, with the same orientation.

Locate the centre

The centre is $(7, 2)$; for example $(7, 3)$ maps to $(7, 5)$ with scale factor 3.

Final answer

Enlargement, scale factor 3, centre $(7, 2)$

Question 25

4MA1-2H Q5 whole

ORIGINAL QUESTION UNIT 2

3 marks

5 Avril bakes a cake.

She uses flour, butter and sugar such that

$$\text{weight of flour} : \text{weight of butter} = 6 : 5$$
$$\text{weight of butter} : \text{weight of sugar} = 3 : 2$$

Avril uses 120 grams of sugar.

Work out the weight of flour Avril uses.

.... grams

Worked solution 25

Chained ratio of flour, butter and sugar

MODULAR UNIT 2
3 marks

whole – Chained ratio of flour, butter and sugar

Find the butter amount

$$\text{Butter:sugar} = 3 : 2, \quad 120 \div 2 = 60, \quad \text{butter} = 3(60) = 180\text{ g}.$$

Use the flour:butter ratio

$$\frac{\text{flour}}{180} = \frac{6}{5} \Rightarrow \text{flour} = \frac{6}{5}(180) = 216\text{ g}.$$

Final answer

216 grams

Question 26

4MA1-2H Q7 whole

ORIGINAL QUESTION UNIT 2

3 marks

7 Hermione buys a boat for \$26 800
The value of the boat depreciates by 8% each year.

Work out the value of the boat at the end of 3 years.
Give your answer correct to the nearest dollar.

\$..

Worked solution 26

Compound depreciation

MODULAR UNIT 2
3 marks

whole – Compound depreciation

Use the depreciation multiplier

$$\text{Each year multiplies the value by } 1 - 0.08 = 0.92.$$

Apply it for three years

$$V = 26800(0.92)^3 = 20868.8384.$$

Round to dollars

$$V \approx \$20\,869.$$

Final answer

\$20,869

Question 27

4MA1-2H Q8 whole

ORIGINAL QUESTION UNIT 2

3 marks

8 The mean number of goals scored by a hockey team in 8 matches is 6
The team plays 2 more matches and scores k goals in each match.
The mean number of goals scored by the hockey team in the 10 matches is 7

Work out the value of k

k = ...

Worked solution 27

Mean after two further matches

MODULAR UNIT 2
3 marks

whole – Mean after two further matches

Find the known totals

$$\text{First total} = 8(6) = 48, \qquad \text{new total} = 10(7) = 70.$$

Find the two new goals

$$2k = 70 - 48 = 22.$$

Solve

$$k = 11.$$

Final answer

k = 11

Question 28

4MA1-2H Q13 whole

ORIGINAL QUESTION UNIT 2

2 marks

13 Robert asked 11 people how many meetings they attended last week.

Here are the results in numerical order.

1 2 4 6 6 8 11 12 13 14 17

Find the interquartile range of the number of meetings.

....

Worked solution 28

Interquartile range from eleven values

MODULAR UNIT 2
2 marks

whole – Interquartile range from eleven values

Read Q1 and Q3

$$Q_1 = 4 \text{ (middle of the lower five)}, \qquad Q_3 = 13 \text{ (middle of the upper five)}.$$

Subtract

$$\text{IQR} = Q_3 - Q_1 = 13 - 4 = 9.$$

Final answer

9

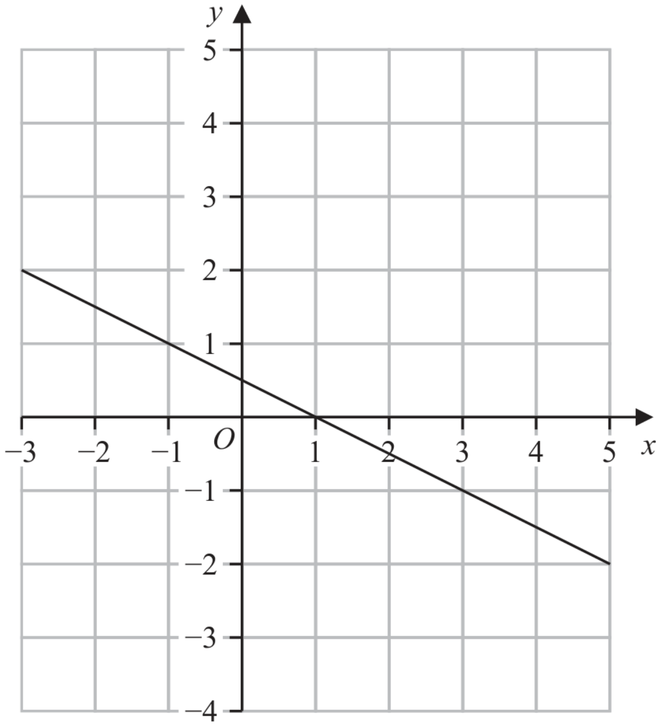
Question 29

4MA1-2H Q14 whole

ORIGINAL QUESTION UNIT 2

3 marks

14 Here is the graph of the equation $2y + x = 1$ drawn on a grid.



By drawing another straight line on the grid, solve the simultaneous equations

$$y - x - 2 = 0$$
$$2y + x = 1$$

$x =$ ....

$y =$ ....

Worked solution 29

Graphical solution of simultaneous equations

MODULAR UNIT 2
3 marks

whole – Graphical solution of simultaneous equations

Draw the second line

$$y - x - 2 = 0 \Rightarrow y = x + 2;\ \text{draw it through } (-2, 0), (0, 2), (2, 4).$$

Read the intersection

$$\text{The two drawn lines meet at } (-1, 1).$$

Final answer

x = -1, y = 1

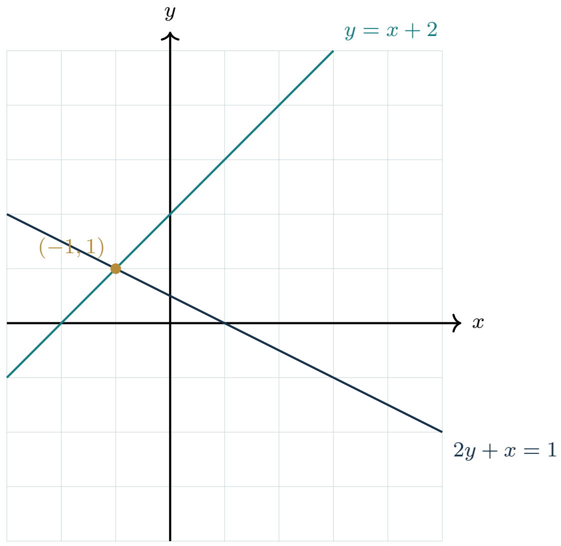
Completed diagram 29

Graphical solution of simultaneous equations

MODULAR UNIT 2

3 marks

Completed solution diagram: Graphical intersection



Question 30
4MA1-2H Q18 whole

ORIGINAL QUESTION UNIT 2
4 marks

18 A particle is moving along a straight line that passes through the fixed point O
The displacement, s metres, of the particle from O at time t seconds is given by

$$s = 2t^3 - 5t^2 + 6t - 5$$

Find the value of t when the acceleration of the particle is 5 m/s²

$t =$ ....

Worked solution 30

Acceleration from a displacement function

MODULAR UNIT 2
4 marks

whole – Acceleration from a displacement function

Differentiate once

$$v = \frac{ds}{dt} = 6t^2 - 10t + 6.$$

Differentiate again

$$a = \frac{dv}{dt} = 12t - 10.$$

Set the required acceleration

$$12t - 10 = 5 \Rightarrow 12t = 15.$$

Solve

$$t = \frac{15}{12} = 1.25 \text{ s}.$$

Final answer

t = 1.25 s

Question 31

4MA1-2H Q19 (a)

ORIGINAL QUESTION UNIT 2

1 marks

19 The functions f and g are such that

$$\mathrm{f}: x \mapsto 5x + 7$$

$$\mathrm{g}: x \mapsto \frac{5}{2x - 9}$$

(a) State which value of x cannot be included in any domain of g

....

(1)

Worked solution 31

Excluded value from a rational function domain

MODULAR UNIT 2

1 marks

(a) – Excluded value from a rational function domain

Set the denominator to zero

$$2x - 9 = 0 \Rightarrow x = \frac{9}{2} = 4.5.$$

Final answer

x = 4.5 cannot be included

Question 32

4MA1-2H Q19 (b)

ORIGINAL QUESTION UNIT 2

2 marks

19 The functions f and g are such that

$$\mathrm{f}: x \mapsto 5x + 7$$

$$\mathrm{g}: x \mapsto \frac{5}{2x - 9}$$

(b) Find fg(4)

....

(2)

Worked solution 32

Composite function value

MODULAR UNIT 2
2 marks

(b) – Composite function value

Evaluate g(4)

$$g(4) = \frac{5}{2(4) - 9} = -5.$$

Apply f

$$fg(4) = f(-5) = 5(-5) + 7 = -18.$$

Final answer

-18

Question 33

4MA1-2H Q19 (c)

ORIGINAL QUESTION UNIT 2

4 marks

The function h is such that

$$h : x \mapsto 3x^2 - 12x + 8 \quad \text{where } x > 2$$

(c) Express the inverse function h^{-1} in the form $h^{-1} : x \mapsto \ldots$

$h^{-1} : x \mapsto$ ....

(4)

Worked solution 33

Find the inverse of a restricted quadratic

MODULAR UNIT 2
4 marks

(c) – Find the inverse of a restricted quadratic

Complete the square

$$y = 3x^2 - 12x + 8 = 3(x-2)^2 - 4.$$

Rearrange

$$y + 4 = 3(x-2)^2 \Rightarrow (x-2)^2 = \frac{y+4}{3}.$$

Use the restriction

$$x > 2 \Rightarrow x - 2 > 0, \text{ so take the positive square root.}$$

Swap variables

$$h^{-1}(x) = 2 + \sqrt{\frac{x+4}{3}}.$$

Final answer

h^-1(x) = 2 + sqrt((x+4)/3)

Question 34

4MA1-2H Q21 whole

ORIGINAL QUESTION UNIT 2

5 marks

21 The line with equation $x + 2y = 5$ intersects the curve with equation $x^2 + 3y^2 = 13$ at the points A and B

Find the coordinates of A and the coordinates of B
Show clear algebraic working.

(.............................. , ...)

(.............................. , ...)

Worked solution 34

Intersections of a line and an ellipse-type curve

MODULAR UNIT 2
5 marks

whole – Intersections of a line and an ellipse-type curve

Substitute the line equation

$$x = 5 - 2y \Rightarrow (5 - 2y)^2 + 3y^2 = 13.$$

Form a quadratic

$$7y^2 - 20y + 12 = 0 = (7y - 6)(y - 2).$$

Solve for y

$$y = \frac{6}{7} \text{ or } y = 2.$$

Recover x

$$y = \frac{6}{7} \Rightarrow x = \frac{23}{7}; \qquad y = 2 \Rightarrow x = 1.$$

State both points

$$(1, 2) \text{ and } \left(\frac{23}{7}, \frac{6}{7}\right).$$

Final answer

$(1, 2)$ and $(23/7, 6/7)$

Question 35

4MA1-2H Q22 (a)(i), (a)(ii), (b)

ORIGINAL QUESTION UNIT 2
5 marks

22

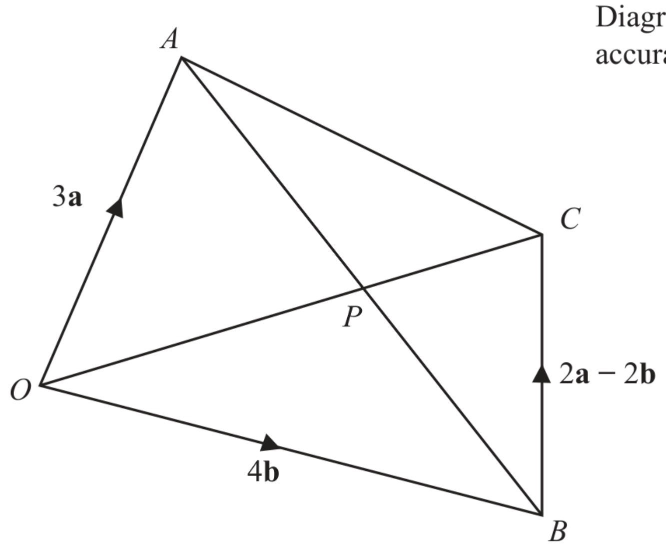


Diagram **NOT** accurately drawn

OACB is a quadrilateral.

$\overrightarrow{OA} = 3\mathbf{a}$ $\overrightarrow{OB} = 4\mathbf{b}$ $\overrightarrow{BC} = 2\mathbf{a} - 2\mathbf{b}$

(a) (i) Find the vector $\overrightarrow{OC}$ in terms of **a** and **b**
Simplify your answer.

$\overrightarrow{OC}$ = ...

(1)

(ii) Find the vector $\overrightarrow{AB}$ in terms of **a** and **b**

$\overrightarrow{AB}$ = ...

(1)

The point *P* lies on *AB* and on *OC*

(b) Using a vector method, find the ratio *AP* : *PB*
Show your working clearly.

....

(3)

Worked solution 35

MODULAR UNIT 2

Find OC from two vectors / Find AB from position vectors / Ratio where two vector descriptions meet **5 marks**

(a)(i) – Find OC from two vectors

Use the path O-B-C

$$\overrightarrow{OC} = \overrightarrow{OB} + \overrightarrow{BC} = 4b + (2a - 2b) = 2a + 2b.$$

Final answer

OC = 2a + 2b

(a)(ii) – Find AB from position vectors

Subtract position vectors

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = 4b - 3a.$$

Final answer

AB = 4b - 3a

(b) – Ratio where two vector descriptions meet

Describe P on both lines

$$\overrightarrow{OP} = \lambda(2a + 2b) = 3a + \mu(4b - 3a).$$

Compare coefficients

$$2\lambda = 4\mu, \qquad 2\lambda = 3 - 3\mu \Rightarrow \mu = \frac{3}{7}.$$

Convert the parameter to a ratio

$$AP : PB = \mu : (1 - \mu) = \frac{3}{7} : \frac{4}{7} = 3 : 4.$$

Final answer

AP:PB = 3:4

Question 36

4MA1-2H Q23 whole

ORIGINAL QUESTION UNIT 2

5 marks

23 Here is a frustum of a cone.
The frustum is made by removing a small cone from a similar large cone.

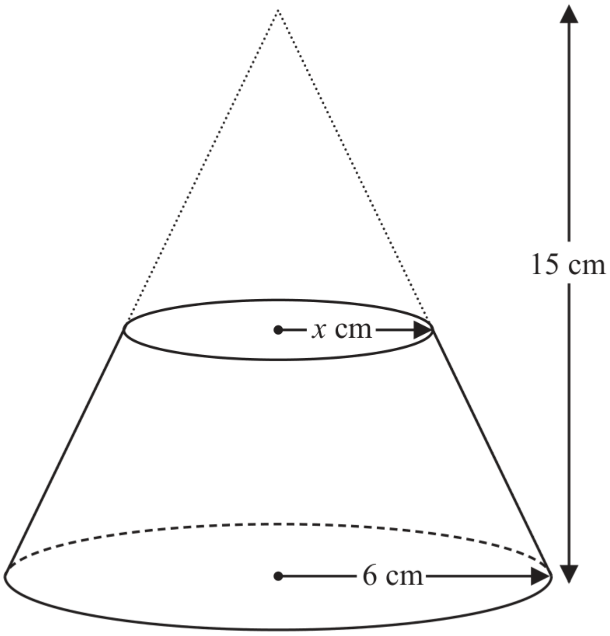


Diagram **NOT** accurately drawn

The height of the large cone is 15 cm.

The radius of the base of the large cone is 6 cm.
The radius of the base of the small cone is x cm.

Given that the volume of the frustum is $\frac{4212}{25}\pi$ cm^3

work out the value of x
Show clear algebraic working.

$x =$ ....

Worked solution 36

Radius of a similar-cone frustum

MODULAR UNIT 2
5 marks

whole – Radius of a similar-cone frustum

Find the large-cone volume

$$V_L = \frac{1}{3}\pi(6^2)(15) = 180\pi.$$

Use similarity

$$\text{The small cone has height } \frac{x}{6}(15) = \frac{5}{2}x, \text{ so its volume is } \frac{1}{3}\pi x^2\left(\frac{5}{2}x\right) = \frac{5}{6}\pi x^3.$$

Use the frustum volume

$$180\pi - \frac{5}{6}\pi x^3 = \frac{4212}{25}\pi \Rightarrow \frac{5}{6}x^3 = \frac{288}{25}.$$

Solve the cubic

$$x^3 = \frac{1728}{125} = \left(\frac{12}{5}\right)^3 \Rightarrow x = \frac{12}{5} = 2.4 \text{ cm}.$$

Final answer

x = 2.4 cm